

Lesson 2.2: Subgroups

Learning Outcomes: At the end of the lesson, the students are able to:

- perform various subgroup tests to determine whether a given subset of a group is a subgroup or not.
- establish properties of subgroups; and
- compute the elements of the subgroups generated by a subset.

The Subgroup Tests

In the previous lesson, we see how subsets inherit the properties of a given set. For instance, the operation of standard addition on the group of real numbers $(\mathbb{R}, +)$ is also a well-defined binary operation on the subset of integers $\mathbb{Z}$, turning $(\mathbb{Z}, +)$ into a group in its own right. This scenario motivates the following definition.

Definition 2.2.1 (Subgroup)

Let G be a group with identity e . A nonempty subset $H \subseteq G$ is called a **subgroup** of G if the following conditions are satisfied:

H1. $e \in H$.

H2. $a, b \in H$ implies $ab \in H$.

H3. $a \in H$ implies $a^{-1} \in H$.

If H is a subgroup of G , we write $H \leq G$.

Remark 2.2.2 If $H \leq G$, then H itself is a group under the operation on G .

To see this, property (H2) guarantees that the binary operation on G acts as a well-defined binary operation in H . Since G is a group, it follows that this operation is inherently associative as a binary operation in H . By, property (H1), the identity element of G acts as a binary operation in H . Finally, by property (H3), every element of H is invertible making H a group in its own right. One checks that H is a subgroup



of G by testing if each of these conditions in **Definition 2.2.1** are satisfied. This is called as the *subgroup test*.

Note that if G is a group, there $G \leq G$ and $\{e\} \leq G$. The subgroup G is called the *improper subgroup*, while the subgroup $\{e\}$ is called as the *trivial subgroup*.

Example 2.2.3 Under the standard number addition, we have $\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}$. On the other hand, under the standard number multiplication, $\mathbb{Q}^\times \leq \mathbb{R}^\times \leq \mathbb{C}^\times$.

Note, however, that $\mathbb{Z}^\times$ is not a subgroup of $\mathbb{Q}^\times$ since for instance, $2 \in \mathbb{Z}^\times$, but 2^{-1} does not exist in $\mathbb{Z}^\times$.

Example 2.2.4 Let H denote the set of all even integers. Prove that $H \leq \mathbb{Z}$ under addition.

Solution. We apply the subgroup test to argue that $H \leq \mathbb{Z}$.

- Note that 0 is even, so $0 \in H$.
- Let $n, m \in H$. Then $n = 2k$ and $m = 2\ell$, for some $k, \ell \in \mathbb{Z}$. Thus,

$$n + m = 2k + 2\ell = 2(k + \ell) \in H.$$

- Let $n \in H$. Then $n = 2k$, for some $k \in \mathbb{Z}$. Thus,

$$-n = -(2k) = 2(-k) \in H.$$

Therefore, $H \leq \mathbb{Z}$.

Note that **Example 2.2.4** may be generalized as follows. The student is asked to provide the details.

Remark 2.2.5 Let $n > 1$ be a given integer and let

$$n\mathbb{Z} := \{nk \mid k \in \mathbb{Z}\}$$

denote the set of all integer multiples of n . Then $n\mathbb{Z} \leq \mathbb{Z}$.

Example 2.2.6 Let G be a group and let $a \in G$ be given. Let

$$C(a) = \{x \in G \mid ax = xa\}$$



be the set of all elements in G which commutes with a . Show that $C(a) \leq G$.

Proof. First, note that $ae = a = ea$. This shows $e \in C(a)$.

Next, we show that $C(a)$ is closed under multiplication. Let $x, y \in C(a)$. Then $ax = xa$ and $ay = ya$. Thus,

$$a(xy) = (ax)y = (xa)y = x(ay) = x(ya) = (xy)a.$$

Hence, $xy \in C(a)$.

Finally, we show that $C(a)$ is closed under inversion. Let $x \in C(a)$. Then

$$ax^{-1} = ax^{-1}a^{-1}a = a(ax)^{-1}a = a(xa)^{-1}a = aa^{-1}x^{-1}a = x^{-1}a.$$

This proves that x^{-1} commutes with a , so $x^{-1} \in C(a)$.

Therefore, by the subgroup test, $C(a) \leq G$. Note that this subgroup is called as the *centralizer* of a in G . $\square$

So far, we have been utilizing a subgroup test which uses three steps. The following result gives us an alternative subgroup test which will require a single step.

Theorem 2.2.7 (One-Step Subgroup Test)

Let H be a nonempty subset of a group G . Then

$$H \leq G \text{ if and only if for all } x, y \in H, xy^{-1} \in H.$$

Proof. Assume that $H \leq G$. Let $x, y \in H$. By property (H3), $y^{-1} \in H$. Thus, by property (H2), $xy^{-1} \in H$.

Conversely, assume that for all $x, y \in H$, it is true that $xy^{-1} \in H$. We show that $H \leq G$ by satisfying properties (H1), (H2), and (H3).

By hypothesis, H is nonempty. Pick any $h \in H$. Take $x = y = h$. By assumption,

$$e = hh^{-1} = ab^{-1} \in H.$$

Thus, property (H1) holds.

Next, let $a \in H$. Take $x = e$ and $y = a$. Then $x, y \in H$. By assumption, we have



$$a^{-1} = e \ a^{-1} = xy^{-1} \in H.$$

Thus, property (H3) holds.

Finally, we show that property (H2) holds. Let $a, b \in H$. Since we know that (H2) already holds, it follows that $b^{-1} \in H$. Thus, by assumption we have

$$ab = a(b^{-1})^{-1} = xy^{-1} \in H.$$

Hence, property (H2) holds.

Since (H1), (H2), and (H3) holds, it follows that $H \leq G$. □

Note that **Theorem 2.2.7** (One-Step Subgroup Test) is expressed using the multiplicative notation. Hence, we adjust contexts when dealing with particular examples. For instance, in the group $(\mathbb{Z}, +)$, a nonempty subset H is a subgroup of $\mathbb{Z}$ if and only if for all $a, b \in H$, it is true that $a + (-b) \in H$. In other words, H must be closed under *subtraction*.

Example 2.2.8 Let $\mathbb{Q}^\times$ be the multiplicative group of nonzero rational numbers. Let

$$H = \{2^n 3^m \mid n, m \in \mathbb{Z}\}.$$

Show that $H \leq \mathbb{Q}^\times$.

Proof. Trivially, every $H \neq \emptyset$ since $1 = 2^0 3^0 \in H$. Also, one may observe that every number of the form $2^n 3^m$ is a nonzero rational number, whenever n and m are integers, so $H \subseteq \mathbb{Q}^\times$. We employ **Theorem 2.2.7** to prove that $H \leq \mathbb{R}^\times$.

Let $r, s \in H$. Write

$$r = 2^{n_1} 3^{m_1} \quad \text{and} \quad s = 2^{n_2} 3^{m_2},$$

where $n_1, n_2, m_1, m_2 \in \mathbb{Z}$. Then

$$\begin{aligned} rs^{-1} &= (2^{n_1} 3^{m_1})(2^{n_2} 3^{m_2})^{-1} \\ &= \frac{2^{n_1} 3^{m_1}}{2^{n_2} 3^{m_2}} \\ &= 2^{n_1 - n_2} 3^{m_1 - m_2}. \end{aligned}$$



Since $n_1 - n_2 \in \mathbb{Z}$ and $m_1 - m_2 \in \mathbb{Z}$, it follows that $rs^{-1} \in H$. By Theorem 2.2.7, $H \leq \mathbb{Q}^\times$. $\square$

For finite subsets of a group, it is relatively easier to tell if they are subgroups or not.

Theorem 2.2.9 (Finite Subgroup Test)

Let G be a group and H be a finite nonempty subset of G . Then H is a subgroup of G if and only if H is closed under the operation on G .

Proof. Trivially, if H is a subgroup of G , then H is closed under the operation in G .

Conversely, assume that H is a nonempty finite subset of G , which is closed under the operation in G . By Theorem 2.2.7, we only need to show that H is closed under inverses. (Why?)

Let $h \in H$. If $h = e$, then $h^{-1} = h \in H$. Assume that $h \neq e$. We consider the positive integer powers of h , namely $h, h^2, h^3, \dots$. Note that by the closure property of H , each of these powers are elements of H . Since H is finite, these powers are not distinct. So, there exist positive integers $n > m$ such that $h^n = h^m$. Consequently,

$$hh^{n-m-1} = h^{n-m} = h^n(h^m)^{-1} = h^n(h^n)^{-1} = e,$$

so $h^{-1} = h^{n-m-1} \in H$.

Therefore, $H \leq G$. $\square$

Example 2.2.10 Determine which of the following subsets of $\mathbb{Z}_{12}$ is/are subgroup/s.

1. $\{0, 4, 8\}$
2. $\{0, 2, 4\}$
3. $\{2, 3, 5, 8\}$

Solution. By Theorem 2.2.9, we only need to check which of these subgroups is closed under addition modulo 12. By manually checking the sum of every pair of elements, we come up with the following conclusion.

1. The subset $\{0, 4, 8\}$ is a subgroup of $\mathbb{Z}_{12}$. Indeed, by looking at the following table below, we see that the set is closed under addition modulo 12.

+	0	4	8
0	0	4	8
4	4	8	0
8	8	0	4



2. The subset $\{0, 2, 4\}$ is *not* a subgroup. As a counterexample, we note that

$$2 + 4 = 6 \text{ and } 6 \notin \{0, 2, 4\}.$$

3. The subset $\{2, 3, 5, 8\}$ is *not* a subgroup either, since $0 \notin \{2, 3, 5, 8\}$.

Example 2.2.11 Determine all the subgroups of the Klein-4 group $V_4 = \{e, a, b, c\}$.

Solution. To do this, we list down all subsets of V_4 containing e , which is closed under multiplication.

- $\{e\}$ is a subgroup, by default.
- $\{e, a\}$ is a subgroup, since $a^2 = a \circ a = e$.
- $\{e, b\}$ is a subgroup, since $b^2 = b \circ b = e$.
- $\{e, c\}$ is a subgroup, since $c^2 = c \circ c = e$.
- The subsets $\{e, a, b\}$, $\{e, a, c\}$, and $\{e, b, c\}$ are not subgroups, since $a \circ b = c$, $a \circ c = b$, and $b \circ c = a$.
- $V_4 = \{e, a, b, c\}$ is the improper subgroup.

Therefore, only $\{e\}$, $\{e, a\}$, $\{e, b\}$, $\{e, c\}$ and V_4 are the subgroups of V_4 . In group theory, whenever a group possesses a finite number of subgroups, we often draw a *lattice diagram of subgroups* which arranges the subgroups from the one with the largest order down to the one with the smallest order, and connect two subgroups by a line, if one is a subset of the other. In the case of V_4 , the lattice diagram of subgroups is shown in **Figure 5**

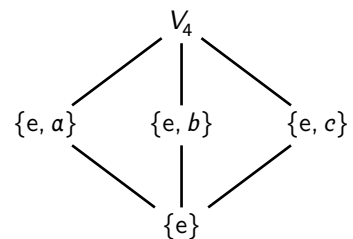


Figure 5: Lattice Diagram of Subgroups for V_4

We next look at how subgroups of a given group interact with one another in terms of set operations.

Theorem 2.2.12 (Intersection Theorem)

Let $\{H_i \mid i \in I\}$ be a collection of subgroups of a group G . Then



$$\bigcap_{i \in I} H_i = \{h \in H_i \mid \text{for all } i \in I\}$$

is also a subgroup of G .

Proof. First of all, note that each H_i is a subgroup of G by hypothesis. Thus, $e \in H_i$ for all $i \in I$. It follows that $e \in \bigcap_{i \in I} H_i$ showing that $\bigcap_{i \in I} H_i \neq \emptyset$.

We now apply **Theorem 2.2.7** to prove that $\bigcap_{i \in I} H_i$ is a subgroup of G . Let $x, y \in \bigcap_{i \in I} H_i$. Then $x, y \in H_i$, for all $i \in I$. Since H_i is a subgroup for each i , it follows that $xy^{-1} \in H_i$, for all $i \in I$. Consequently,

$$xy^{-1} \in \bigcap_{i \in I} H_i.$$

Therefore, $\bigcap_{i \in I} H_i$ is a subgroup of G by **Theorem 2.2.7**. □

If the index set $I = \{1, 2, \dots, n\}$ is finite, **Theorem 2.2.12** guarantees that the intersection

$$H_1 \cap H_2 \cap \dots \cap H_n \leq G$$

whenever $H_i \leq G$, for $i = 1, 2, \dots, n$. However, note that the index I in the theorem is arbitrary. Meaning, even when we have infinitely many subgroups in consideration, we can be guaranteed that their intersection is also a subgroup.

Example 2.2.13 Consider the subgroups $6\mathbb{Z}$ and $8\mathbb{Z}$ of the additive group $\mathbb{Z}$.

- $6\mathbb{Z} = \{0, \pm 6, \pm 12, \pm 18, \pm 24, \pm 30, \pm 36, \pm 48, \dots\}$
- $8\mathbb{Z} = \{0, \pm 8, \pm 16, \pm 24, \pm 32, \pm 40, \pm 48, \pm 56, \dots\}$

implying that $6\mathbb{Z} \cap 8\mathbb{Z} = \{0, \pm 24, \pm 48, \pm 72, \dots\} = 24\mathbb{Z}$. The students are asked to convince themselves that this is because $24 = \text{lcm}(6, 8)$.

Note 2.2.14 In general,

$$m_1\mathbb{Z} \cap m_2\mathbb{Z} \cap \dots \cap m_k\mathbb{Z} = \ell\mathbb{Z} \quad \text{if and only if} \quad \ell = \text{lcm}(m_1, m_2, \dots, m_k).$$



Warning

The union of two subgroups **need not** be a subgroup.

For example, if we consider the subgroups $2\mathbb{Z}$ and $3\mathbb{Z}$ of the group $\mathbb{Z}$, the union $2\mathbb{Z} \cup 3\mathbb{Z}$ is not a subgroup since

$$2, 3 \in 2\mathbb{Z} \cup 3\mathbb{Z}, \quad \text{but } 2 + 3 = 5 \notin 2\mathbb{Z} \cup 3\mathbb{Z}$$

Subgroups Generated by a Subset

We have seen in the previous section that not every subset is a subgroup. For instance, let's take an arbitrary subset S of a group G . Note that S need not be a subgroup. If we consider the collection

$$\mathcal{C} = \{H \leq G \mid S \subseteq H\},$$

then the elements of $\mathcal{C}$ are the subgroups of G that contains S as a subset. If we take the intersection of the subgroups that are in the collection $\mathcal{C}$, we see that

$$\bigcap_{H \in \mathcal{C}} H \leq G$$

by [Theorem 2.2.12](#), and consequently, this intersection also contains S as a subset. Easily put, $\bigcap_{H \in \mathcal{C}} H \in \mathcal{C}$. In terms of set inclusion, the subgroup $\bigcap_{H \in \mathcal{C}} H$ is the *smallest* subgroup of G containing S as a subset, that is,

$$\text{if } K \leq G \text{ and } S \subseteq K, \text{ then } \bigcap_{H \in \mathcal{C}} H \subseteq K.$$

Definition 2.2.15 Let S be a subset of a group G . The **subgroup generated by** S is defined as the intersection of all subgroups of G that contain S as a subset. This subgroup is denoted by $\langle S \rangle$.

Remark 2.2.16 In a group G ,

1. $\langle \emptyset \rangle = \{e\}$.
2. Due to subgroup properties (H1), (H2), (H3), the subgroup $\langle S \rangle$ contains all the finite products of integer powers of the elements of S .



Remark 2.2.16 (2) implies that $\langle S \rangle = \{s_1^{n_1} s_2^{n_2} \dots s_k^{n_k} \mid s_i \in S, \text{ where } k, n_k \in \mathbb{Z}, k > 0\}$. For instance, if $a, b \in S$, then $\langle S \rangle$ must contain all the following products:

$$ab^{-1}a^3b^{-27}, \quad a^{-2}bab^2, \quad a^{-1}b^3a^{41}bab^3, \quad \text{etc.}$$

For brevity of notation, if $S = \{s_1, s_2, \dots, s_k\}$ is a finite set, we simply write

$$\langle S \rangle = \langle s_1, s_2, \dots, s_k \rangle.$$

Example 2.2.17 Consider the set $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$ and the group $S(\mathbb{R}^\times)$ of bijections from $\mathbb{R}^\times$ to $\mathbb{R}^\times$ under function composition. Let $f : \mathbb{R}^\times \rightarrow \mathbb{R}^\times$ and $g : \mathbb{R}^\times \rightarrow \mathbb{R}^\times$ be functions defined by

$$f(x) = -x, \quad g(x) = \frac{1}{x}, \quad \text{for } x \in \mathbb{R}^\times.$$

Note that f and g are bijections, so $f, g \in S(\mathbb{R}^\times)$. We aim to find $\langle f, g \rangle$. Consider the following computations for $x \in \mathbb{R}^\times$:

- $f^2(x) = (f \circ f)(x) = f(f(x)) = f(-x) = -(-x) = x$. It follows that $f^2 = \text{id}_{\mathbb{R}^\times}$. Thus, f is a group element of order 2. It follows that

$$f^n = \begin{cases} \text{id}_{\mathbb{R}^\times}, & \text{if } n \text{ is even} \\ f, & \text{if } n \text{ is odd.} \end{cases}$$

- $g^2(x) = (g \circ g)(x) = g(g(x)) = g\left(\frac{1}{x}\right) = 1/(1/x) = x$. Thus, $g^2 = \text{id}_{\mathbb{R}^\times}$. Similarly, g is a group element of order 2 and

$$g^n = \begin{cases} \text{id}_{\mathbb{R}^\times}, & \text{if } n \text{ is even} \\ g, & \text{if } n \text{ is odd.} \end{cases}$$

- $(f \circ g)(x) = f(g(x)) = f(1/x) = -\frac{1}{x}$ and $(g \circ f)(x) = g(f(x)) = g(-x) = -\frac{1}{x}$. Thus, $f \circ g = g \circ f$.

From these computations, we can deduce that

$$\langle f, g \rangle = \{\text{id}_{\mathbb{R}^\times}, f, g, f \circ g\}.$$

Additionally, we can check that this set is already closed under composition of functions, hence, it is a subgroup by **Theorem 2.2.9**.

$\circ$	$\text{id}_{\mathbb{R}^\times}$	f	g	$f \circ g$
$\text{id}_{\mathbb{R}^\times}$	$\text{id}_{\mathbb{R}^\times}$	f	g	$f \circ g$
f	f	$\text{id}_{\mathbb{R}^\times}$	$f \circ g$	g
g	g	$f \circ g$	$\text{id}_{\mathbb{R}^\times}$	f
$f \circ g$	$f \circ g$	g	f	$\text{id}_{\mathbb{R}^\times}$



Definition 2.2.15 allowed us *salvage* a given subset S of a group G when it is not a subgroup by constructing the subgroup $\langle S \rangle$. In other words, the definition guaranteed that there exists a smallest subgroup which contains S . In other words, we have produced a method that generates a subgroup out of the subset S . The case when S is a singleton is not an exception.

Definition 2.2.18 (Cyclic Subgroup)

Let G be a group and $a \in G$. The **cyclic subgroup generated by a** is defined as the subgroup

$$\begin{aligned}\langle a \rangle &= \{a^n \mid n \in \mathbb{Z}\} \\ &= \{e, a, a^{-1}, a^2, a^{-2}, \dots\}.\end{aligned}$$

The cyclic subgroup generated by a consists of all the integer powers of a . In the additive notation, however, the subgroup $\langle a \rangle$ consists of all the integer multiples of a , namely, $e, a, -a, 2a, -2a, \dots$

Example 2.2.19 The cyclic subgroups of $\mathbb{Z}$ are the subgroups $\langle 0 \rangle = \{0\}$ and

$$\langle n \rangle = \{nk \mid k \in \mathbb{Z}\} = n\mathbb{Z},$$

where n is a positive integer. Note that in this group, $\langle n \rangle = \langle -n \rangle$.

Example 2.2.20 List down the distinct cyclic subgroups of $\mathbb{Z}_6$.

Solution. To compute the cyclic subgroups of $\mathbb{Z}_6$ under addition modulo 6, we calculate successive multiples ka for each element a until we reach the identity element 0. (Why?)

- For $a = 0$:
 - $1a = 0$. $\therefore \langle 0 \rangle = \{0\}$.
- For $a = 1$:
 - $1a = 1$;
 - $2a = 1 + 1 = 2$;
 - $3a = 1 + 1 + 1 = 3$;



- $4a = 1 + 1 + 1 + 1 = 4$;
- $5a = 1 + 1 + 1 + 1 + 1 = 5$;
- $6a = 1 + 1 + 1 + 1 + 1 + 1 = 6 \equiv 0 \pmod{6}$.

$$\therefore \langle 1 \rangle = \{1, 2, 3, 4, 5, 0\} = \mathbb{Z}_6.$$

- For $a = 2$:

- $1a = 2$;
- $2a = 2 + 2 = 4$;
- $3a = 2 + 2 + 2 = 6 \equiv 0 \pmod{6}$.

$$\therefore \langle 2 \rangle = \{2, 4, 0\}.$$

- For $a = 3$:

- $1a = 3$;
- $2a = 3 + 3 = 6 \equiv 0 \pmod{6}$.

$$\therefore \langle 3 \rangle = \{3, 0\}.$$

- For $a = 4$:

- $1a = 4$;
- $2a = 4 + 4 = 8 \equiv 2 \pmod{6}$;
- $3a = 4 + 4 + 4 = 12 \equiv 0 \pmod{6}$.

$$\therefore \langle 4 \rangle = \{4, 2, 0\}.$$

- For $a = 5$:

- $1a = 5$;
- $2a = 5 + 5 = 10 \equiv 4 \pmod{6}$;
- $3a = 5 + 5 + 5 = 15 \equiv 3 \pmod{6}$;
- $4a = 5 + 5 + 5 + 5 = 20 \equiv 2 \pmod{6}$;
- $5a = 5 + 5 + 5 + 5 + 5 = 25 \equiv 1 \pmod{6}$;
- $6a = 5 + 5 + 5 + 5 + 5 + 5 = 30 \equiv 0 \pmod{6}$.

$$\therefore \langle 5 \rangle = \{5, 4, 3, 2, 1, 0\} = \mathbb{Z}_6.$$

These computations imply that $\mathbb{Z}_6$ has exactly four distinct cyclic subgroups:

- $\langle 0 \rangle = \{0\}$
- $\langle 1 \rangle = \mathbb{Z}_6 = \langle 5 \rangle$



- $\langle 2 \rangle = \{0, 2, 4\} = \langle 4 \rangle$
- $\langle 3 \rangle = \{0, 3\}$

Example 2.2.21 Consider the multiplicative group $\mathbb{C}^\times$ of nonzero complex numbers. Consider the element $z = i = \sqrt{-1}$. Note that

- $z^1 = i$
- $z^2 = i \cdot i = \sqrt{-1} \cdot \sqrt{-1} = -1$
- $z^3 = i^2 \cdot i = (-1) \cdot i = -i$
- $z^4 = i^3 \cdot i = (-i) \cdot i = -i^2 = -(-1) = 1.$

Since 1 is the identity element of $\mathbb{C}^\times$, it follows that the cyclic subgroup generated by i is

$$\langle i \rangle = \{1, i, -1, -i\}.$$

On the other hand, if we consider the complex number $w = 2i$, we see that

$$w^n = 2^n i^n, \quad \text{for all } n \in \mathbb{Z}.$$

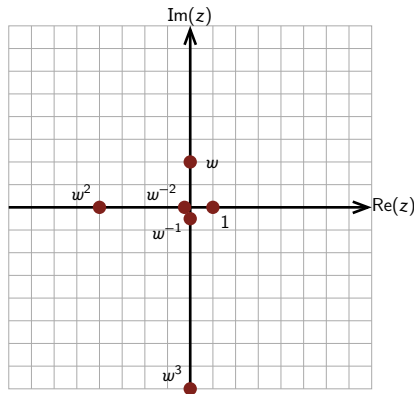
Some values of w^n are summarized in [Table 9](#).

n	-2	-1	0	1	2	3
w^n	$-\frac{1}{4}$	$-\frac{1}{2}i$	1	$2i$	-4	$-8i$

Table 9: Values of $(2i)^n$

Since $2^n \rightarrow \infty$ as $n \rightarrow \infty$, we can deduce that w^n has infinitely many values. In fact, if the values in [Table 9](#) are plotted in the complex plane, we can see a spiral pattern as in [Figure 6](#).



Figure 6: Values of w^n

As we have observed, the cyclic subgroup $\langle a \rangle$ can be finite or infinite depending on the choice of element a . The following theorem gives a complete characterization.

Theorem 2.2.22 Let G be a group and let $a \in G$.

1. If a has an infinite order, then

$$a^k = a^\ell \text{ if and only if } k = \ell.$$

Consequently, $\langle a \rangle = \{e, a, a^{-1}, a^2, a^{-2}, \dots\}$ is infinite.

2. If a has a finite order n , then

$$a^k = a^\ell \text{ if and only if } k \equiv \ell \pmod{n}.$$

Consequently, $\langle a \rangle = \{e, a, a^2, \dots, a^{n-1}\}$.

Proof.

1. Assume that a has an infinite order. Then $a^n = e$ if and only if $n = 0$. Thus,

$$\begin{aligned} a^k = a^\ell &\iff (a^k)(a^\ell)^{-1} = e \\ &\iff a^{k-\ell} = e \\ &\iff k - \ell = 0 \\ &\iff k = \ell. \end{aligned}$$

It follows that $k \neq \ell$ implies $a^k \neq a^\ell$. Consequently,



$$\langle a \rangle = \{e, a, a^{-1}, a^2, a^{-2}, \dots\}$$

has distinct elements, hence, infinite.

2. Assume that $|a|_G = n$. Then $a^k = e$ if and only if n divides k . Thus,

$$\begin{aligned} a^k = a^\ell &\iff (a^k)(a^\ell)^{-1} = e \\ &\iff a^{k-\ell} = e \\ &\iff n \mid k - \ell \\ &\iff k \equiv \ell \pmod{n}. \end{aligned}$$

Finally, the inclusion $\{e, a, a^2, \dots, a^{n-1}\} \subseteq \langle a \rangle$ is trivially true as $\langle a \rangle$ contains all the integer powers of a , by definition. To prove the reverse inclusion, let $x \in \langle a \rangle$. Then $x = a^m$ for some $m \in \mathbb{Z}$. By the division algorithm, there exist $q, r \in \mathbb{Z}$ such that

$$n = mq + r, \quad \text{where } 0 \leq r < n.$$

Hence,

$$\begin{aligned} a^m &= a^{nq+r} = (a^n)^q a^r \\ &= e^q a^r = a^r. \end{aligned}$$

The condition that $0 \leq r < n$ implies that $r \in \{0, 1, \dots, n-1\}$. Therefore,

$$x = a^m = a^r \in \{e, a, a^2, \dots, a^{n-1}\},$$

proving that $\langle a \rangle \subseteq \{e, a, a^2, \dots, a^{n-1}\}$. □

The following is a consequence of **Theorem 2.2.22**.

Corollary 2.2.22.1 Let G be a group and $a \in G$. Then $|a|_G = |\langle a \rangle|$.

Let $m \in \mathbb{Z}$ be given. In the case when a has a finite order n and if r is the remainder when m is divided by n , then $m \equiv r \pmod{n}$. **Theorem 2.2.22** (2) guarantees that

$$a^m = a^r.$$



Corollary 2.2.22.2 If a is a group element of order n , then for any integer m , $a^m = a^r$, where r is the remainder when m is divided by n .

Example 2.2.23 Given the complex number $i = \sqrt{-1}$. Compute the product

$$(2 + i)(2 + i^2)(2 + i^3) \cdots (2 + i^{100}).$$

Solution. Note that i has a group order 4 under complex number multiplication. Hence, the value of i^k , cycles every 4 terms.

- If $k \equiv 0 \pmod{4}$, then $2 + i^k = 2 + i^0 = 3$.
- If $k \equiv 1 \pmod{4}$, then $2 + i^k = 2 + i$.
- If $k \equiv 2 \pmod{4}$, then $2 + i^k = 2 + i^2 = 2 + (-1) = 1$.
- If $k \equiv 3 \pmod{4}$, then $2 + i^k = 2 + i^3 = 2 - i$.

Thus, the product of the first four terms is

$$3 \cdot (2 + i) \cdot 1 \cdot (2 - i) = 15.$$

Therefore, because $100 = 4 \cdot 25$, we have

$$(2 + i)(2 + i^2)(2 + i^3) \cdots (2 + i^{100}) = \underbrace{15 \cdot 15 \cdots 15}_{25 \text{ times}} = 15^{25}.$$

